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TOPICAL REVIEW

Machine Learning and Deep Learning Approaches for Accent Recognition: A Review

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ABSTRACT Accent recognition has attracted immense research interest owing to the advancements in automatic speech recognition (ASR) systems. Accent variations are an essential factor in speech, and can drastically decrease the performance of ASR systems. This study presents an extensive examination of various accent-recognition models. In this paper, we explain various preprocessing techniques, different feature extraction methods, and a detailed methodology based on the machine learning (ML) and deep learning (DL) approaches used for accent recognition. Papers were selected from Google Scholar, IEEE Xplore, ACM Digital Library, ScienceDirect, Springer, Research Gate, Scopus, and the Directory of Open Access. The search returned 103 papers, including (journals and conferences) covering the scope of this study from 2015 to 2023. Detailed discussions on the various characteristics of audio signals are presented for the development of an accent recognition system. This paper provides an overview of various DL techniques to highlight the current state-of-the-art approaches, along with the associated activation and loss functions. It also offers a concise discussion on commonly used datasets for accent recognition systems and examines the challenges posed by different auditory characteristics. Finally, we discuss the research gaps, ideas for improvement, and the current and future trends proposed by different authors.

INDEX TERMS Accent recognition, automatic speech recognition, deep learning, machine learning, preprocessing, feature extraction.

I. INTRODUCTION

The recent advancements in ML and DL have radically transformed speech-recognition applications with remarkable outcomes. Research and development in speech recognition systems is attempting to understand human speakers in different environments with diverse accents and languages. Depending on acoustic features, such as accent of speaking, sex, size of the vocal tract, gender, emotions, and others, each person's speech signals differ from those of the other, which allows researchers and scientists to distinguish different speakers. In recent years, speech recognition has attained exceptional improvement with advancements in DL models and expendable open-access data. An accent is a feature of the speech of a person, characterized by a systematic difference in pronunciation patterns as a result of distinct cultural

backgrounds and ethnolinguistics. Accent recognition is an essential step in ASR as it degrades the performance of these systems [1].

It is a challenging task to recognize the correct accent followed by transferring it to a speech recognizer that has been trained for a certain accent group. Accent detection is itself a complicated task, as accents are distinguished by phoneme pronunciation, voice quality, and prosody. Accent detection has recently garnered more interest worldwide as a result of the incredible growth of speech recognition technologies made available by telephone networks and computers. Furthermore, it will always be difficult for a computer to replicate the human capacity to recognize speech as one of the most influential causes of speech/speaker variability other than health conditions, gender, speaking rate, mood states, and so forth. It is always difficult for a machine to mimic the human capacity to recognize speech. Automatic accent identification is a difficult issue for speech technology

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and it is speculated that using real conversational speech may be beneficial for many sorts of paralinguistic tasks [2].

The accent recognition system consists of three components, as indicated in Figure 1 preprocessing module, extraction of feature module, and recognition model. Through accent detection, a system of intelligent investigation can monitor suspected criminal actions and enable surveillance investigations of suspects who keep their identities hidden on purpose. Criminals use phone calls and voice messages to commit various offenses. This technology can recognize them by their accent and bring them to trial [3], [4]. Additionally, a person's accent or dialect may reveal information regarding their origin and culture [5]. Non-native accents can be detected in tourism and travel centers and automated international telephone call systems. If the listener recognizes the dialect, he/she can locate an appropriate assistant who speaks the client's native language [6].

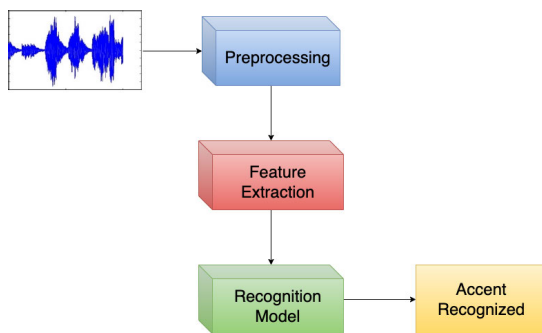


FIGURE 1. Block diagram of accent recognition.

Accent recognition includes the use of databases, various processes such as preprocessing, feature extraction techniques, recognition approaches (classical, ML, and DL), and evaluation processes (accuracy, precision, recall, F1-measure, equal error rate). Again, the various steps in this process have an independent significance. The initial phase in accent detection is preprocessing, in which we filter the speech signal to remove muteness, followed by pre-emphasis, framing, windowing, and finally normalization. Figure 2 shows the categorization of the accent recognition system. It provides a complete overview of the requirements and potential approaches for accent identification.

The various abbreviations used in this study are listed in Table 1. The remainder of this paper is organized as follows. Section II explains the research questions on which the review is about, and follows with the strategy for the selection of papers and investigation. Section III explains the detailed preprocessing and feature extraction strategies. Section IV provides an in-depth explanation of ML techniques. Section V discusses the DL methods, activation functions, loss functions, and optimizers used in the various methods, their performance, and different evaluation metrics for accent recognition. Section VI provides an overview of accent recognition in various Indian languages. Future trends and research directions in the field of accent recognition

are discussed in Section VII. Finally, the conclusions are discussed in Section VIII.

II. RESEARCH QUESTIONS AND PAPER SELECTION

This study provides a comprehensive analysis of various ML and DL-based accent classification methods. All the reported literature is systematically shortlisted, and the distribution of year-wise papers is shown in Figure 3. The major findings of this study are discussed based on the following research questions. Articles relevant to the queries were scientifically selected by adding the following search terms:

- RQ1: What are the major preprocessing methods for accent recognition?
- RQ2: What are the most significant feature extraction methods for accent recognition?
- RQ3: What are the different classical and ML methods and techniques applied for accent recognition?
- RQ4: What are the different DL methods and techniques applied for accent recognition?
- RQ5: What are the future trends and research gaps in accent Recognition?

The literature survey was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. The PRISMA flow diagram is presented in Figure 4. The PRISMA diagram outlines four key stages: Identification, which represents the number of records retrieved from databases and other sources; Screening, where studies are assessed based on their title and abstract, and duplicates are removed; Eligibility, which includes the selection of full-text articles based on their in-depth experimental procedures; and Inclusion, which represents the final number of studies incorporated into the review. This systematic approach ensures transparency and clarity in the selection process, facilitating a comprehensive and well-structured analysis. To acquire an overview of the papers on accent recognition, the search was carried out by searching for journal and conference articles in four databases: Scopus, IEEE Xplore Digital Library, Science Direct, and ACM Digital Library. The search results were collected and saved using the Mendeley Reference Management Software. As shown in Figure 5, the criteria for the exclusion and inclusion of papers in the research were established. The inclusion criteria were as follows:

- Important keywords (e.g., accent recognition, accent classification, and accent detection).
- Sorted by the title.
- Duplicates removed by the Mendeley
- Sorted papers (journal, conference) for final analysis.

III. PREPROCESSING AND FEATURE EXTRACTION TECHNIQUES FOR ACCENT RECOGNITION

RQ1: What are the major preprocessing methods for accent recognition?

Preprocessing of audio speech is critical for the construction of an effective accent-identification system. VAD

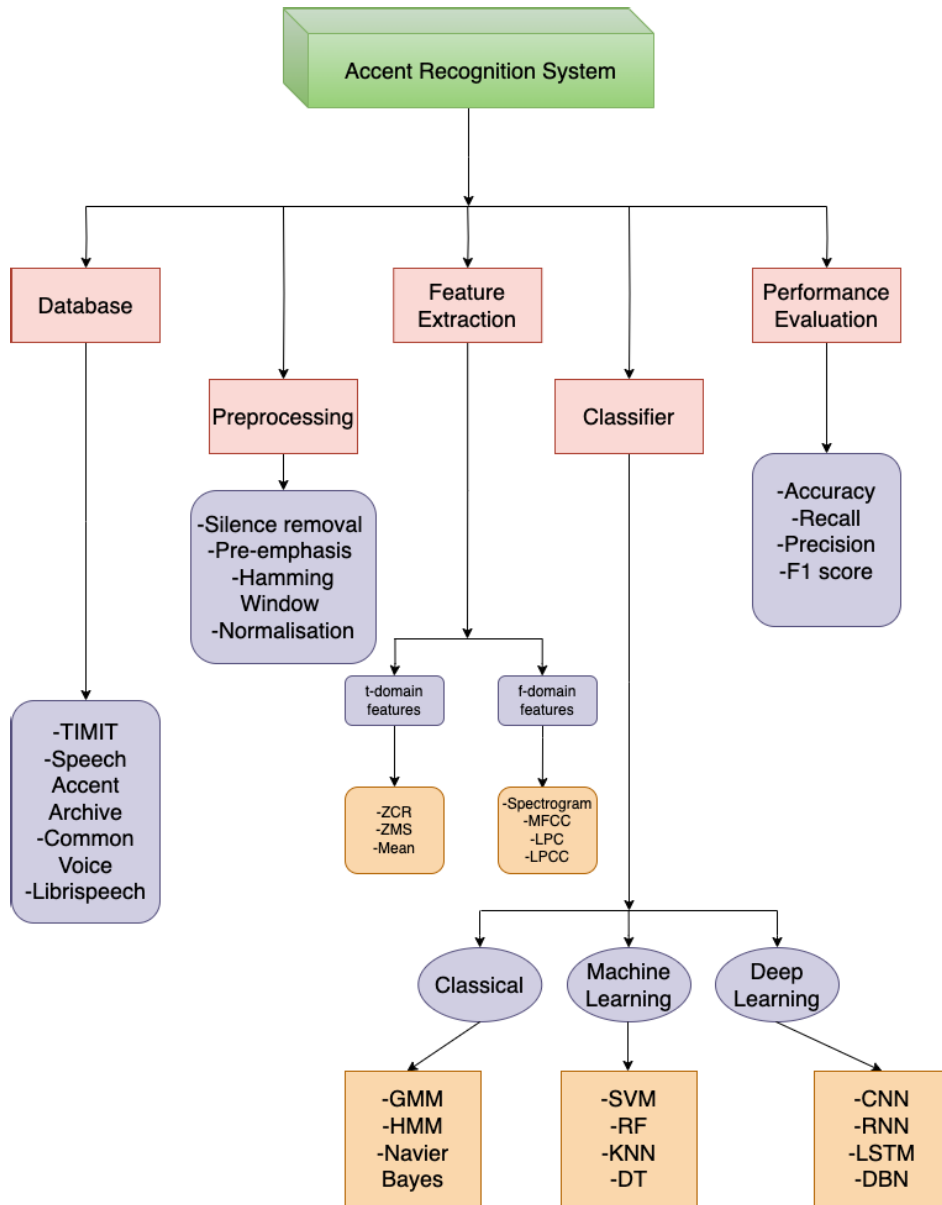


FIGURE 2. Classification of accent recognition system.

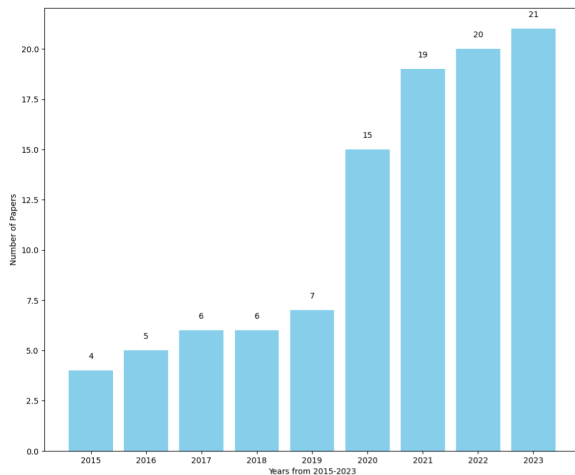
are devices that separate a speech signal into unvoiced voiced chunks as well as speech and non-speech components. Voice activity detection has been an effective method for the preprocessing of speech signals for many years. Noise cancellation, pre-emphasis, and dimensionality reduction of speech make the system more computationally efficient. According to the aforementioned studies, the silence removal preprocessing strategy is preferred by the majority of studies published between 2015 and 2023 to extract the appropriate information for accent detection in the presence of additive noise. Others have attempted to ensure compatibility with the feature extraction approach using techniques such as pre-emphasis and hamming windows. The various preprocessing techniques can be summarized as follows:

A. PREPROCESSING TECHNIQUES IN ACCENT RECOGNITION

Numerous preprocessing strategies have been used by researchers in various accent-recognition fields. Extracting adequate information for accent identification in the presence of additive noise is challenging. Therefore, the first and most important step is to reduce the distortion in the voice stream to improve system performance. To ensure compatibility with the feature extraction method, this unnecessary component must be eliminated during preprocessing. Table 2 presents the key preprocessing strategies utilized in accent identification, highlighting their respective benefits and drawbacks, along with relevant references for further insights.

TABLE 1. List of abbreviations with definitions (sorted in ascending order).

Abbreviation	Definition	Abbreviation	Definition
AESRC	Accented English Speech Recognition Challenge	AE	Autoencoder
Adam	Adaptive Moment Estimation	ANN	Artificial Neural Network
ASR	Automatic Speech Recognition	BLSTM	Bidirectional Long Short Term Memory
CER	Character Error Rate	CNN	Convolutional Neural Network
CTC	Connectionist Temporal Classification	DBN	Deep Belief Network
DCT	Discrete Cosine Transform	DL	Deep Learning
DNN	Deep Neural Network	DT	Decision Tree
DT-CWPT	Dual-Tree Complex Wavelet Packet Transform	DWT	Dynamic Time Warping
EER	Equal Error Rate	ELM	Extreme Learning Machine
F0	Fundamental Frequency	F-BANK	Filter Bank
FAR	False Acceptance Rate	FCNN	Fully Convolutional Neural Network
FRR	False Rejection Rate	GAN	Generative Adversarial Network
GB	Gradient Boosting	Ge-LU	Gaussian Error Linear Unit
GMM	Gaussian Mixture Model	H-ELM	Hierarchical Extreme Learning Machine
HMM	Hidden Markov Model	JFA	Joint Factor Analysis
K-NN	K-Nearest Neighbour	LEAF	A Learnable Frontend for Audio Classification
LFBE	Log Filter Bank Energies	LFCC	Linear Frequency Cepstral Coefficients
LPCC	Linear Predictor Cepstral Coefficients	LPF	Low Pass Filter
MFCC	Mel-Frequency Cepstral Coefficients	PSD	Power Spectral Density
LSTM	Long Short Term Memory	MAE	Mean Absolute Error
MFDWC	Mel Frequency Discrete Wavelet Coefficient	ML	Machine Learning
ML-ELM	Multilayer Extreme Learning Machine	MLP	Multi-Layer Perceptron
MSE	Mean Square Error	NNC	Nearest Neighborhood Classifier
PCA	Principal Component Analysis	PLP	Perceptual Linear Prediction
PSO	Particle Swarm Optimization	Re-LU	Rectified Linear Unit
ReMKMFCC	Renyi Entropy-Based Multi-Kernel Mel Frequency Cepstral Coefficient	Res-Nets	Residual Neural Network
RF	Random Forest	RMSprop	Root Mean Squared Propagation
RMSE	Root Mean Square Error	RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic	SDC	Shifted Delta Cepstral
SCG	Scaled Conjugate Gradient	SNR	Signal-to-Noise Ratio
SGD	Stochastic Gradient Descent	STFT	Short Term Fourier Transform
SSC	Spectral Subband Centroid	SVM	Support Vector Machine
TDNN	Time-Delay Neural Network	UBM	Universal Background Model
VAD	Voice Activity Detection	WER	Word Error Rate
VQ	Vector Quantization	WPT	Wavelet Packet Transform
ZCR	Zero Crossing Ratio	XAI	Explainable AI

**FIGURE 3.** Year-wise paper distribution.

1) SILENCE REMOVAL

A technique for reducing the processing time, thereby improving the system efficiency. It also improves overall system performance by removing unnecessary portions of the voice signal. Silence trimming enhances accuracy and efficiency by eliminating unnecessary pauses, reducing processing time, and improving feature extraction for more robust speech modeling [7], [8], [9], [10], [11].

In text-independent acoustic systems, silence trimming is commonly employed using pitch-based VAD, as described in [2]. Due to the presence of non-fluent English speakers, speech samples often contain frequent pauses. To address this, VAD is applied to eliminate silence periods by removing segments exceeding 300 milliseconds while also accounting for background noise [3]. Hassan et al. [12] implemented silence trimming to eliminate unintended pauses, mitigating delays ranging from 0.5 to 1.5 seconds at the beginning of speech and prolonged pauses between sentences during paragraph reading. Mikhailava et al. [13] investigated the influence of pause occurrences in audio files on classification outcomes, demonstrating that pauses serve as strong indicators of foreign accents, particularly in Romance languages compared to Germanic ones. Additionally, several studies have employed the VAD algorithm for silence removal as part of preprocessing [14], [15], [16]. Table 2 summarizes various studies incorporating silence trimming as a preprocessing technique.

2) PRE-EMPHASIS

This is described as the process of increasing the magnitude of a higher frequency concerning a lower frequency to enhance the quality of the signal. The energy of lower-frequency speech signals is increased

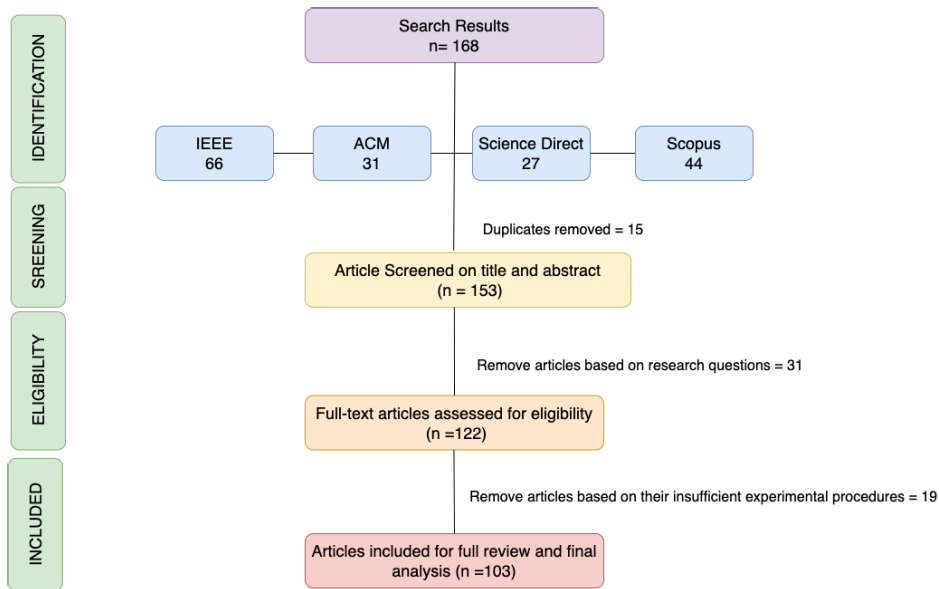


FIGURE 4. PRISMA flow chart.

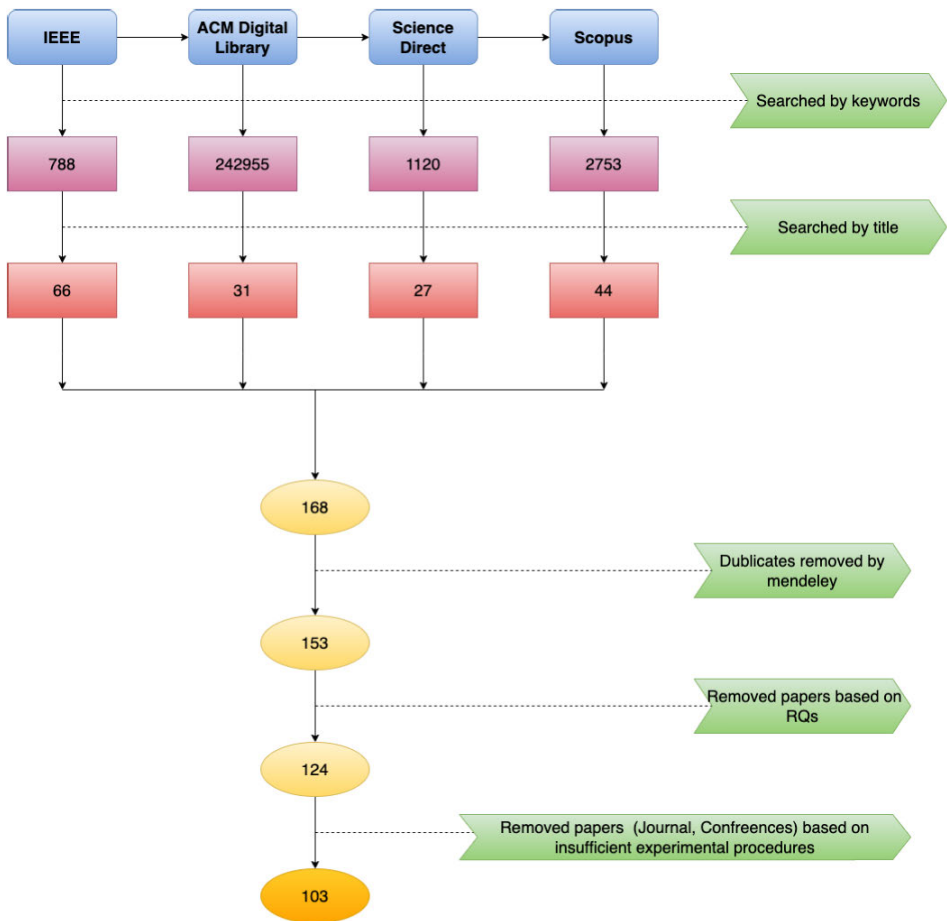


FIGURE 5. Search strategy and selection of papers.

by pre-emphasizing them to make them stronger than higher-frequency noise components, thereby immediately improving the SNR. Reference [17] implemented this

technique for all components with a sound intensity below 30 dB. In accent recognition, pre-emphasis enhances high-frequency components by compensating for spectral

attenuation, thereby improving the signal-to-noise ratio and preserving critical speech features [5], [18], [19]. This process reinforces phoneme distinction, facilitates more accurate feature extraction, and ultimately enhances overall system performance [7], [20], [21].

3) HAMMING WINDOW

In this method, the ideal impulsive response is multiplied by a window function, as shown in Figure 6 [19], which is then used to construct a filter that reduces the ideal impulse response. These methods are commonly utilized in constructing digital filters and converting an infinitely long impulsive response to a finitely long impulse response. This process is commonly referred to as the windowing technique. The Hamming window plays a crucial role in accent recognition by reducing spectral leakage through the smooth tapering of signal edges, thereby preserving essential frequency components [5], [14], [18]. This refinement enhances feature extraction, improves spectral resolution, and ultimately contributes to greater classification accuracy [22].

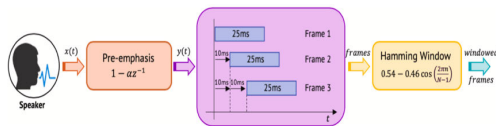


FIGURE 6. Pre-emphasis framing and hamming windowing for sound signal.

4) ENDPOINT/OUTLIER DETECTION

Speech endpoint identification is a method of identifying speech boundaries using digital processing tools. Redundant information in a voice sample is typically obtained before and after the actual speech area. In [23], the authors employed anomaly detection techniques to identify and eliminate malicious data points that could compromise the performance of algorithms. The Z-score method was utilized for outlier detection and removal, demonstrating its effectiveness, particularly for medium-sized datasets.

5) NORMALIZATION

The normalization approach involves executing a linear adjustment of the original data to normalize each feature on the same scale. Normalization of acoustic information or extracted characteristics is a critical stage in accent recognition technology. The study conducted by [13] leveraged crowd-sourced data and implemented normalization on each audio recording prior to model training to mitigate variations caused by linear distortions. Similarly, [24] applied normalization to standardize both input and target data, thereby enhancing the neural network's learning efficiency.

6) SPEECH SEGMENTATION

Audio segmentation separates a digital audio stream into a series of segments or frames, which are then classified

into accent recognition, music, or noise. Audio-signal processing relies heavily on segmentation. The study conducted by [5] employed a segmentation approach, wherein each audio recording, approximately three minutes in duration, was divided into separate.wav files using a VAD system. Similarly, [17] applied audio segmentation followed by exploratory data analysis. Furthermore, [24] implemented segmentation to distinguish voiced segments (vowels), unvoiced segments (consonants), and pauses within the speech dataset.

A review of related studies concludes that preprocessing techniques are meticulously designed to mitigate the impact of noisy environments, thereby enhancing the robustness of accent recognition systems. Techniques such as silence removal and endpoint detection effectively eliminate non-informative segments, reducing computational complexity [25], [34] while simultaneously minimizing the influence of background noise on feature extraction, ultimately improving the performance of deep learning models [21]. VAD identifies speech segments by distinguishing them from non-speech elements; its effectiveness in noisy conditions can be further enhanced through filtering techniques that refine speech isolation [46]. Pre-emphasis amplifies high-frequency components, thereby improving speech clarity, particularly in low-quality recordings [34]. The application of a Hamming window smooths the edges of each frame, reducing spectral leakage and enhancing the precision of feature extraction [47]. Additionally, audio normalization enhances the generalization capability of voice-controlled systems [48], while speech segmentation isolates relevant speech components, minimizing interference from non-speech elements. Collectively, these preprocessing techniques significantly contribute to improving recognition accuracy.

B. FEATURE EXTRACTION METHODS IN ACCENT RECOGNITION

RQ2: What are the most significant feature extraction methods for accent recognition?

According to the literature, the most widely used feature extraction technique is based on MFCC. The extracted feature vectors are primarily derived from standard MFCCs, without the incorporation of additional features [2], [6], [8], [10], [12], [20], [21], [26]. Alternatively, feature vectors have also been constructed by augmenting MFCCs with additional techniques, such as SDC, delta, and double delta coefficients, to enhance the representational capacity of the extracted features [29], [32], [49].

Recent studies on i-vector-based modeling techniques, which utilize DNNs as classifiers for bottleneck feature extraction, have demonstrated superior performance compared to traditional approaches relying solely on MFCC-based feature extraction [9]. A comprehensive analysis of the existing literature indicates that most accent recognition methodologies predominantly employ ML algorithms in conjunction with MFCC feature extraction. However,

TABLE 2. Different preprocessing techniques for accent recognition.

Methods	Benefits	Drawbacks	References
Silence Removal	Processing time is reduced [25].	It goes well with white Gaussian noise only [16].	[2], [3], [26] , [27], [12], [13], [14], [18], [15], [28], [29], [21], [7], [8], [9], [10], [11], [30], [31], [32], [33]
Pre-emphasis	Enhances negative spectral slopes of voice segments, improving performance [34].	Low frequency gain increases as well [35].	[5], [18], [20], [21], [17], [7], [9], [36], [37], [19]
Hamming Window	Decreases inaccuracies in distortion estimation.	Huge variance in spectral estimation.	[5], [14], [18], [15], [38], [27], [39], [40], [22]
EndPoint/Outlier Detection	Includes spectral data while minimizing computational resources [34].	Frequently fails in low SNR environments [41].	[23]
Audio Normalization	Prevents estimation inaccuracy due to changes in speaker volume [34].	Jitter magnitude increases (affecting precision) [41].	[13], [42], [36], [24], [28], [43], [39], [44]
Speech Segmentation	Helps to recognize individuals more accurately [17].	Increases computational cost and may lead to loss of contextual information [5].	[5], [17], [24], [45]

there has been no significant increase in studies utilizing pure MFCC techniques without the integration of additional features [13], [17], [23], [50], [51].

The feature-extraction approach is typically used for raw signals to recover significant information that may contain additive noise. The collected acoustic characteristics must provide sufficient data to properly match the accent identification models. Background noise, fluctuation in the utterance of speech by the speaker owing to sickness or emotional condition, noise in the surroundings, the noise present in the transmission channel, and so on all affect the speech quality and accuracy of accent detection. To overcome this problem and extract proper characteristics, a feature extraction procedure for accent identification is required. Consequently, an adequate feature extraction approach should be capable of extracting features in real-time, while retaining as much information as possible. Speech characteristics, such as temporal and spectral features, can also be used to classify feature extraction techniques.

The time domain is commonly utilized in temporal analysis techniques to examine speech signals in their raw form [24]. As the name suggests, spectral analysis represents the speech signal in the frequency domain, enabling a more detailed investigation of its spectral characteristics [3], [17], [23], [42], [51], [52]. Several studies have explored both temporal features—such as rhythm, fundamental frequency, and pitch—as well as spectral features—including spectral entropy, spectral flatness, spectral centroid, and spectral flux—to extract relevant characteristics for accent recognition in speech [23], [36]. In [53], the authors investigated speaker profiling using short-term spectral features of English

vowels for accent classification among Indian speakers. Their study compared various feature computation techniques, revealing that spectral features outperformed cepstral ones, with lower frequency filters playing a pivotal role. The findings underscored the necessity of balancing time window duration and frequency resolution to achieve optimal classification, with potential applications in forensic speech analysis.

This research investigated vocal and prosodic features along with speaker embeddings for accent identification in a diverse English speech dataset. A hierarchical classification framework was designed, employing a bottom-up strategy to distinguish between native and non-native accents [54]. A separate study demonstrated that identifying native versus non-native English speech based solely on visual cues achieved an accuracy of 76.5%, even under noisy or missing audio conditions, using HMMs [55]. Furthermore, [56] proposed a Tamil accent identification system leveraging MFCCs and visual features (optical flow). A hybrid sequential CNN-HMM model was employed to classify accents from three regions of Tamil Nadu, achieving an accuracy of 96.7% through multi-feature fusion, outperforming methods based on individual features. Similarly, [57] investigated Hindi dialect classification using a combination of spectral (MFCC) and prosodic features. A neural network trained on 48,000 utterances attained an accuracy of 82%, highlighting the effectiveness of feature fusion in dialect recognition.

Table 3 presents the feature extraction techniques employed by various authors in this study and examines different aspects of widely recognized feature extractors, highlighting their advantages and limitations in accent identification. Among these techniques, MFCC remains the

most powerful and extensively utilized feature extraction method. The literature predominantly explores MFCC as a standalone feature for accent recognition. Additionally, it is frequently combined with other features, such as spectral roll-off, ZCR, spectrogram, chromagram, and spectral centroid, to enhance performance. The block diagram depicting the MFCC feature extraction process from the raw input signal is illustrated in Figure 7. As a feature extraction approach, MFCC aims to capture essential characteristics from an audio signal for subsequent processing.

Consequently, MFCC continues to be the principal feature extraction technique in accent recognition, either as an independent method or in conjunction with supplementary features such as delta coefficients and spectral attributes. While traditional MFCC-based approaches have demonstrated efficacy, recent advancements integrating DNNs and hybrid models have yielded superior performance, particularly in accent classification tasks requiring enhanced robustness. The existing literature emphasizes the necessity of optimizing feature extraction methodologies to improve classification accuracy while addressing challenges associated with background noise and speaker variability.

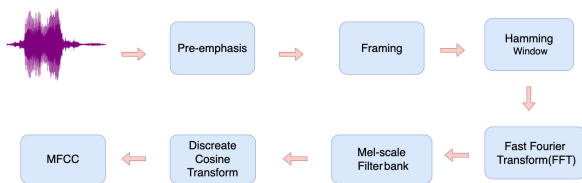


FIGURE 7. Block diagram of feature extraction of MFCC from audio signal.

IV. CLASSICAL AND MACHINE LEARNING METHODS FOR ACCENT RECOGNITION

RQ3: What are the different classical and ML methods and techniques applied for accent recognition?

The GMM, KNN, and SVM appear to be the most common choices for detecting the accent of a person. Their capacity to accurately simulate the temporal information available in speech signals is the reason for their widespread use. The authors explored various models for accent recognition in speech, including SVM [5], [42], KNN [60], [61], and RF [29], [44]. Among these, GMM hybrid approaches, particularly GMM-UBM, demonstrated strong performance in accent classification, achieving an accuracy of 90.9% [14], whereas standalone GMM attained 87.9%. Another hybrid model, GMM-SVM, proposed in [32], exhibited a competitive accuracy of 80%. In contrast, conventional GMM-based approaches [20], [30] yielded relatively moderate results, indicating the limitations of GMM in effectively capturing accent variations.

ANN [21] outperformed traditional classifiers such as SVM, KNN, and RF, demonstrating superior learning capabilities. Furthermore, DL-based hybrid models, such as the DNN-HMM [97], achieved notable improvements in accent classification. These advanced models leverage the intricate

associations between speech audio characteristics and their corresponding accents, as evidenced in [11] and [39].

The choice of features and models significantly impacts the performance of traditional and machine learning-based accent recognition systems. Different feature sets, such as MFCCs, filter banks, pitch, and prosodic features, capture various aspects of speech, influencing the classification accuracy of these models. Additionally, different models vary in their effectiveness at capturing temporal and spatial dependencies in speech data. In the study by [52] on accent recognition, the ELM classifier achieved an accuracy of 95.28% for English numerical speech by leveraging a combination of wavelet and MFCC features. For Malay words, the accuracy improved to 96.72% by incorporating wavelet, LPC, and MFCC-based feature extraction techniques. Similarly, [36] conducted a comparative analysis of feature extraction methods for Thai accent recognition across three regional dialects. The study evaluated energy spectral density, PSD, MFCC, and spectrogram-based features, utilizing an SVM with a radial basis function (RBF) kernel. The results demonstrated that MFCC outperformed other techniques, achieving an accuracy of 94.9%.

A classification model was developed using classical algorithms such as GMM [20], GMM-UBM [29], and ML techniques including DT [23], SVM [45], and ELM [22] for training data. A comparative analysis by [21] evaluated the performance of various ML models, revealing that the ANN classifier achieved the highest accuracy (88.53%), followed by KNN (86.11%), SVM (85.54%), and RF (85.28%). The developed model demonstrated its capability to estimate the speaker's accent from unknown utterances. This study introduces Grad-Transfer, a novel descriptor derived from Gradient-weighted Class Activation Mapping for CNN interpretability. Grad-Transfer enables accent classification by transferring CNN-extracted knowledge to traditional ML models. Experiments on the Voice Cloning Toolkit dataset show that Grad-Transfer enhances macro average accuracy and unweighted recall by up to 23.00% and 23.58%, respectively, outperforming spectrogram-based models [98].

Furthermore, experiments using GMM-UBM [87] and i-vector representations on the FarsDat dataset successfully identified five native accents. The results indicated that articulatory features outperformed spectral features, such as MFCC and SDC features. Additionally, PCA applied to phoneme attributes reduced error rates by 45.12%, yielding an accuracy of 75.29% and enhancing robustness in low-data conditions. Several classical and ML-based techniques (Table 4) have been explored for accent recognition, including DT, GMM, SVM, RF, and hybrid approaches like GMM-SVM. Various researchers in this domain have investigated these methods, highlighting their advantages and limitations in accent identification.

Thus, the effectiveness of accent recognition systems depends on the choice of both feature extraction techniques and classification models. Traditional classifiers such as GMM, SVM, and KNN have demonstrated

TABLE 3. Benefits and drawbacks of feature extraction techniques for accent recognition.

Methods	Benefits	Drawbacks	References
MFCC	• Recognizes phonemes in the speech by detecting phonatory objects similarly to the human ear [27]. • Uses a spectrum to reflect phoneme utterance, suitable for speech recognition [51]. • Reduces undesirable spectral fluctuations in higher frequency ranges [14].	• Not noise-resistant [26]. • Highest accuracies achieved with native speaker data [58]. • Unable to extract voice signal features at lower frequencies [6].	[2], [6], [26], [12], [21], [8], [10], [20], [37], [39], [22], [44], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [14], [73], [27], [43], [74], [45], [75], [76], [32], [29], [49], [36], [77]
LPCC	• Uses cepstral analysis to eliminate strong correlation of LPC [78]. • More resilient than straight-forward LPC analysis.	• May not adequately depict speech as it assumes the signal is stationary. • Unable to retain previous knowledge during testing.	[73], [79]
Spectrogram	• Provides equal bandwidth and similarly spaced frequency components [58]. • Better at representing contextual differences in speech [77].	• Poor time and frequency resolution [41].	, [80], [81], [45], [75], [76], [58], [38], [36], [77]
ZCR	• Ideal for retrieving music information, e.g., distinguishing voiced and unvoiced signals [23].	• Noise sensitive; may produce incorrect results in noisy environments.	[43], [50], [13], [51]
PLP	• Performs well in hybrid settings [14]. • Computationally efficient [82]	• Spectral balance can be influenced by noise, channel, and recording equipment. • Does not perform well with clean voice signals.	[30], [31], [14].
Spectral Centroid	• Performs well in noisy environments [51].	• Not effective for music genre classification.	[43], [50], [13]
SSC + MFDWC	• Provides finer details of accent. • Performs well on both noisy and clean data [82], [83].	• May increase computational cost.	
LPC	• Accurate and dependable, it reflects vocal tract characteristics [84]. • Resilient and effective with low-bit-rate voice inputs.	• Feature coefficients are highly correlated. • Cannot distinguish between words with similar-sounding phonemes [85]	[14], [73], [74], [5], [42].
SDC	• Captures language-specific spectral features for long-term temporal information [11].	• Performs poorly in noisy environments.	[73], [11], [86], [29], [49], [87]
Discrete Wavelet Transform	• Considers both temporal and frequency information. • Effective for de-noising tasks [88].	• Inflexible as all signals are processed with the same base wavelet.	
Filter bank	• Mimics the human ear’s frequency response for better feature extraction.	• Discards phase information, which is useful in certain tasks.	[89], [90], [48], [91], [92], [93], [94], [68], [95], [96]
Formant Frequencies	• Captures key resonant frequencies of the vocal tract for speech characterization.	• Sensitive to speaker variability and may require additional normalization for robustness.	[5]

moderate to strong performance, with hybrid approaches like GMM-UBM and GMM-SVM further enhancing accuracy.

However, deep learning-based models, particularly ANN, and hybrid architectures like DNN-HMM, have significantly

TABLE 4. Classical and ML techniques for accent recognition in reviewed study.

Methods	Benefits	Drawbacks	References
GMM	Requires fewer parameters for learning, which can be correctly approximated via expectation maximization.	Requires a large amount of data for accurate accent classification.	[14], [30], [33], [20], [37], [60], [65], [70], [74]
HMM	- Capable of handling distortion and noise in voice signals as well as changes in speaker, accent, and environment. - Simple to implement. - Capable of modeling both discrete and continuous sequences.	- Performance relies strongly on selected characteristics and parameters, which may impact results. - Assumes the Markov property.	[6], [7], [55]
GMM-UBM	Outperforms ordinary GMM in terms of performance [99].	Increases model complexity.	[2], [14], [15], [29], [8], [11], [31], [32], [39], [65], [73], [87]
GMM-SVM	Performs better than standard GMM and SVM [100].	Takes longer to train.	[2], [32], [39]
DT	Quick for predicting unknown records and simple to create; ignore undesirable qualities.	- Easily overfits; large DTs may be difficult to interpret. - Small differences in training data can lead to imbalances in decisions.	[5], [45], [23], [43]
SVM	- Distinguishes between native and non-native categories [60]. - Less susceptible to overfitting and provides higher classification accuracy [82].	- Performs poorly when features exceed training samples; unsuitable for massive databases. - Computational cost increases with the number of output classes.	[5], [42], [15], [21], [36], [22], [52], [60], [61], [45], [27], [82], [7], [67], [40], [73], [72]
KNN	- Simple classifier that bases classification on the strength of the training dataset [82]. - Resistant to noisy training datasets [82].	- Unsuitable for high-dimensional and categorical features due to difficulty in calculating distances. - High classification costs for large datasets.	[5], [42], [21], [20], [52], [60], [61], [66], [82], [101]
DWT	Effective in detecting similarities and differences between waveforms of speech utterances [26].	Computationally expensive and slow.	[26], [69]
RF	With its fast speed and cheap computing cost, the RF classifier is a useful method for dealing with overfitting and imbalance problems [17].	- More trees are required for increased accuracy. - More trees delay the model down.	[17], [29], [44], [45], [72]
ELM	- ELMs are appealing for accent detection tasks because they can be trained fast and give superior generalization capacity with little quantities of training data [22]. - ELMs require less time to [22].	- Its accuracy may not always be as high as that of other DL models. - The number of hidden nodes affects ELM.	[42], [40], [22], [52], [87]
i-vector	- When compared to Joint factor analysis, the computation cost is substantially lower [102]. - For recognizing foreign accents, the i-vector technique outperforms the traditional GMM-UBM [103].	- i-vector training is dependent on the data selection. - For short utterances, the i-vector technique performs poorly.	[39], [8], [11], [32], [65], [68], [73], [86], [49]

outperformed classical methods by capturing intricate temporal and spectral dependencies in speech. Additionally, feature selection plays a crucial role in improving classification accuracy, with MFCC, wavelet, and Chroma features showing notable success. The introduction of advanced techniques, such as Grad-Transfer and I-vector representations, has further refined accent recognition, making models more robust in real-world applications.

V. DEEP LEARNING METHODS FOR ACCENT RECOGNITION

RQ4: What are the different DL methods and techniques applied for accent recognition?

This section reviews various DL frameworks applied to accent recognition and discusses their strengths and weaknesses. It examines the impact of different loss functions, activation functions, and optimizers on the model performance. In addition, this section highlights the evaluation metrics used to assess accuracy and effectiveness, along with the diverse datasets employed in accent recognition research, providing a comprehensive overview of the methodologies and tools utilized in this domain.

A. ACCENT RECOGNITION USING DEEP LEARNING

Over the past decade, DL approaches have made great progress in numerous ML applications, and they currently substitute nearly all state-of-the-art ML methods in diverse disciplines. DL approaches have demonstrated superior performance over classical ML techniques in accent and speech detection, such as the use of MFCC features with GMMs [104]. The primary advantage of DL methods lies in their ability to extract highly abstract and discriminative embedding features directly from speech signals. Several standalone DL architectures, including CNNs [62], LSTM networks [64], and BLSTM networks [95], have been explored for accent recognition. Additionally, hybrid architectures such as CNN-LSTM [38] and pre-trained models like ResNets [45] have been developed to enhance feature extraction and classification accuracy. Researchers have further investigated hybrid approaches that integrate conventional ML models with DL-based architectures, such as DNN-HMM [97], CNN-SVM [81], and transformer-based models [91], to improve speaker accent recognition.

Because the basic MLP has been (and continues to be) an ANN's core structure, the conventional training method is the SGD search. Advancements in fundamental approaches to accent recognition have been significantly driven by increased processing power and the availability of large datasets [105]. DL models, including DNNs, LSTM networks, and DBNs, have demonstrated promising outcomes in this domain. Several researchers have explored hybrid architectures, such as CNN-LSTM and CNN-SVM, yielding improved performance in accent recognition tasks. Reference [9] achieved a WER of 22.1% by leveraging a BLSTM model combined with CTC loss. Similarly, [45] employed transfer learning as a foundational technique for

low-resource accent identification systems. Reference [90] introduced a transformer-based model and conducted extensive experiments for accent classification, highlighting its effectiveness.

A notable study proposed a 1D CNN-BiGRU-Attention model utilizing F-Bank and MFCC features for English accent recognition, achieving an F1 score of 85.52%. The attention mechanism significantly enhanced classification performance across American, British, and Indian accents [47]. Additionally, [106] investigated Bangla speaker accent classification using ZCR, MFCC, RMS, and Mel-Spectrogram features, attaining 94% accuracy on unseen data. Among the evaluated models, DNNs outperformed LSTM, Stacked LSTM, and deep CNN models, which achieved 67%, 71%, and 85% accuracy, respectively. Reference [96] further explored multilingual accent classification using Channel Attention within TDNNs and Wav2Vec 2.0, achieving an impressive accuracy of 95% on English accents.

A SpeechBrain-based approach utilizing Common Voice 7.0 (English) and Common Voice 11.0 (Italian, German, and Spanish) datasets was introduced, incorporating a multi-task network for joint speech and accent recognition. This framework integrates CTC, accent recognition, and ASR branches, enhancing task-specific learning. By leveraging accent embeddings and aligned text, the model improved recognition accuracy on English and Chinese datasets, demonstrating its robustness [107]. Reference [108] presented an automated Arabic accent identification method based on a speech recognizer trained on phoneme variations, classifying five Arabic accents (Gulf, Iraqi, Egyptian, Levantine, and Maghrebi) with an accuracy range of 79–86%, surpassing previous methods. Furthermore, [109] investigated accent recognition in ASR systems using DL frameworks, specifically CNNs. The study compared MFCC and Chroma-based features, revealing that Chroma features exhibited superior performance in distinguishing Italian, Turkish, and Korean accents.

For Chinese accent recognition, the Chinese accent recognition model utilized Wav2Vec 2.0 and LSTM, achieving 88.6% accuracy on a dataset of non-native Chinese learners from eight countries. This model demonstrated superior performance in distinguishing similar accents, outperforming conventional techniques [110]. Hybrid models continue to play a crucial role in accent recognition, as they integrate complementary architectures such as CNNs and LSTMs to enhance classification accuracy. For instance, employing CNN for non-native English accents and LSTM models resulted in improved accuracy, with CNN achieving 79.69% and LSTM reaching 84.33%, emphasizing the advantage of hybrid approaches in accent classification tasks [38].

Furthermore, [81] demonstrated that a CNN-SVM hybrid model significantly enhanced accent recognition accuracy. While SVM alone achieved 88% accuracy and CNN reached 95%, their combined CNN-SVM model achieved the highest performance at 97%, illustrating the effectiveness of hybrid architectures. Similarly, [3] reported that an RNN model achieved an accuracy of 42.9%, a DNN attained 49.1%,

and their combined DNN-RNN model yielded the highest accuracy of 50.2%. In the context of Indian English accents, an ASR framework incorporating speech segment endpoint detection with Mel-Spectrogram features and a CNN-BLSTM model was introduced. Evaluated on a 29-hour dataset, the model achieved a training accuracy of 98.67%, validation accuracy of 93.62%, and an ASR achieved WER of 11.63% [111].

Moreover, advanced architectures such as Conformer have been explored for accented speech recognition [112]. This model integrates general and accent-specific encoders and is trained using a multi-pass approach with cross-information fusion. The Aformer architecture demonstrated significant improvements in ASR for English and Mandarin datasets, reducing error rates by 10.2%–24.5% compared to the standard Conformer baseline. Table 5 provides an overview of various DL methods employed for accent recognition, highlighting their benefits and drawbacks. It summarizes key approaches, including CNNs, LSTMs, and hybrid models, along with their respective advantages, such as improved feature extraction and robustness. Additionally, the table lists studies that have applied these methods, offering insights into their effectiveness and limitations in accent recognition tasks.

Thus, DL has revolutionized accent and speech recognition by outperforming traditional ML techniques through advanced feature extraction and classification methods. Various architectures, including CNNs, LSTMs, and hybrid models like CNN-LSTM and CNN-SVM, have demonstrated significant improvements in accuracy and robustness. The integration of attention mechanisms, transformer-based models, and pre-trained architectures has further enhanced recognition performance across multiple languages and datasets. Recent studies highlight the continued evolution of hybrid approaches and multi-task learning frameworks, underscoring their effectiveness in handling complex accent variations.

B. DIFFERENT LOSS FUNCTIONS, ACTIVATION FUNCTIONS AND OPTIMIZERS FOR ACCENT RECOGNITION

DL-based accent detection models differ primarily in four areas: dataset, network architecture, loss function, and optimizer. The activation function of a node defines the output based on a single input or a collection of inputs. It determines how much that node contributes to the node or the nodes after it. Both nonlinear and linear activation functions are possible. Nonlinear activation functions are generally recommended for DL applications. These can produce more accurate results than the linear activation functions [117].

The loss function determines how well the model can predict the output by mapping the actual output. If the model correctly predicts the result, it outputs a small value; otherwise, it outputs a large value. Different studies have used the MSE, cross-entropy, categorical cross-entropy, negative likelihood loss, and other loss functions for accent identification. An optimizer is a function or algorithm that modifies NN parameters, such as weights and learning rates.

Consequently, it lowers the total loss and enhances the model's accuracy. SCG, RMSprop, Adam, and AdaMax are optimizers used to train the DL model for accent recognition. Table 6 provides a quick description of the activation functions, loss functions, and optimizers applied in the various DL models as well as their performance in accent recognition. Figure 8 illustrates the taxonomy of the various DL methods, loss functions, and activation functions utilized for accent recognition.

C. PERFORMANCE EVALUATION METRIC FOR ACCENT RECOGNITION

System performance measurements are important in the critical examination of classification. Consequently, numerous performance indicators were utilized as classifier assessment criteria. In many applications, the confusion matrix estimate is the most extensively used statistic for evaluating the classifier performance. The frequencies of the evaluation criteria in the papers analyzed are shown in Table 7.

T_{positive} (true positive) denotes cases in which both the predicted and actual labels are accurate (i.e., positive), whereas T_{negative} (true negative) indicates instances in which both the actual and predicted labels are incorrect (i.e., negative). The term F_{negative} (false-negative) refers to situations in which the real label is positive and the projected label is negative. Finally, F_{positive} (false positive) displays examples in which the actual label is negative, but the projected label is positive. Reduce F_{positive} and F_{negative} to obtain optimal accent recognition performance.

Several performance metrics were utilized to evaluate the classifier's performance. Accuracy, recall, precision, and F-measure were the most frequently used metrics for accent recognition. The performance indicators are summarized below.

1) ACCURACY

It is mostly utilized as a performance metric, indicating the number of examples successfully identified by a certain classification method. This provides the proportion of accurately anticipated cases to the total number of instances. The mathematical expression for accuracy is as follows:

$$\text{Accuracy} = \frac{T_{\text{positive}} + T_{\text{negative}}}{T_{\text{positive}} + T_{\text{negative}} + F_{\text{positive}} + F_{\text{negative}}} \quad (1)$$

2) PRECISION

Precision is the proportion of true positive predictions among all positive predictions. The accuracy and F_{positive} rates were inversely linked. This is useful for verifying the accuracy of the classification system as an assessment criterion.

$$\text{Precision} = \frac{T_{\text{positive}}}{T_{\text{positive}} + F_{\text{positive}}} \quad (2)$$

3) RECALL

Recall is the ratio of positive cases in which both the predicted and actual labels are correct or positive, that is, T_{positive} to the

TABLE 5. Benefits and drawbacks of DL techniques for accent recognition.

Methods	Benefits	Drawbacks	References
DNN	- Improved discriminating in confusion zones of diverse accents [72]. - Simple to implement.	- Can only accept fixed-length inputs. - Disregards time information in voice signals.	[3], [72], [68], [79], [70], [72], [106]
DNN-HMM	Simple to use for real-time speech and accent identification [97].	More computationally costly than regular DNN.	[97]
ANN	- Robust in nature. - Self-organizing and self-learning. - Adaptable to new environments.	- Highest accuracies achievable only with limited data from native speakers. - Prone to overfitting and local minima problems.	[21], [20], [24], [43], [63], [7], [87], [74], [71]
TDNN	- Classifies patterns with shift-invariance. - Captures temporal dependence within a specified time range.	Limited to predefined temporal ranges.	[93], [96]
CNN-SVM	- Performs better than individual CNN and SVM models [81]. - Converges quickly and reduces overfitting.	Computationally intensive training.	[81]
CNN	- Facilitates the feature extraction process [58]. - Requires less training time.	- Limited in capturing long-term dependent information due to convolution kernel size. - Inability to accurately capture temporal aspects.	[50], [13], [17], [10], [23], [77], [62], [45], [113], [75], [80], [114], [10], [115], [51], [94]
CNN-LSTM	Delivers better results than individual models [38].	Overfitting is a significant concern in time series data categorization.	[38], [23]
LSTM	- Best method for classification and prediction with time series data [23]. - Addresses the issues of exploding and vanishing gradients.	More complex than ordinary RNNs, requiring more training data.	[28], [59], [23], [64], [106]
BLSTM	- Captures complex connections in the input sequence. - Utilizes both past and future contexts for prediction.	- Slower training time than LSTM. - More computationally expensive.	[68], [9], [95]
DBN	Parameter flexibility allows simulation of complex non-linear interactions necessary for audio recognition [44].	- Requires massive data for better performance. - Slow learning process needing a labeled dataset.	[18], [40], [44], [116]
Transformer	Integrates ideas from various models.	Difficult to train and performs poorly on long sequence lengths.	[90], [91], [114]
ResNets	- Outperforms standard deep neural networks. - Allows for faster convergence and simpler training.	- Prone to overfitting on short datasets. - Difficult to interpret internal workings.	[17], [45]

total number of positive instances. It is also known as the true positive rate.

$$\text{Recall} = \frac{T_{\text{positive}}}{T_{\text{positive}} + F_{\text{negative}}} \quad (3)$$

4) F-MEASURE

The F-measure is the weighted harmonic mean of recall and accuracy when F_{positive} and F_{negative} are in perfect

equilibrium. The typical F-measure is F1, which values recall and accuracy equally

$$\text{F-measure} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

5) EQUAL ERROR RATE

EER is used to calculate the average of the FAR and FRR. The lower the EER number, the greater the accuracy of the

TABLE 6. DL Techniques: Activation function, loss function, optimizer, performance, and hardware requirement for accent recognition.

Method	Activation Function	Loss Function	Optimizer	Performance	Requirements
MLP-ANN [24]	Sigmoid	MSE	SCG	Accuracy: 80.7%	Experiments were performed on Google Colaboratory cloud environment.
CNN [50]	ReLU	Categorical Cross Entropy	Adam	Accuracy: 98%	The lowest execution time is 17 minutes and 57 seconds.
CNN [58]	ReLU	Log Loss	SGD	Accuracy: 93.38%	Intel I79750 h CPU, Samsung Evo Plus 970 SSD, Nvidia GeForce GTX 1650 and 64 GB RAM.
CNN [75]	Leaky ReLU	Cross Entropy	Adam	Accuracy: 97.37%	Ubuntu 22.04 LTS, with AMD Ryzen 5 3500u CPU 3.60 GHz processor and 8 GB RAM.
DNN [68]	ReLU	Cross Entropy	Adam	Average: 68%	Experiments were performed on Google Colaboratory cloud environment.
CNN [13]	ReLU	Categorical Cross Entropy	Adam	Accuracy: 98%	Keras with TensorFlow, Librosa, Scikit-learn, Matplotlib and Comet.ml for tracking and storing results.
CNN (VF-Net) [76]	ReLU	Cross Entropy	Adam	Accuracy: 70%	PyTorch deep learning framework and trained Nvidia GTX 1070 Ti GPU.
DeepSpeech2 [12]	Softmax	Novel Loss Function	SGD	WER: 18.8%	Nvidia's 1080 Ti GPU having 3584 Cuda cores and 11 GB cud memory.
ID-CNN [113]	ReLU	MSE	Adam	Accuracy: 97%	–
CNN [17]	ReLU	Cross Entropy	Adam	Accuracy: 76%	DL TensorFlow framework (2.0) and Keras using Google Colab Pro.
ResNet18 [45]	ReLU	Categorical Cross Entropy	Adam	Accuracy: 99.9%	Experiments were performed on Google Colaboratory cloud environment.
Word2Vec [118]	Leaky ReLU	CTC Loss	Adam	WER: 14%	Utilized the SpeechBrain toolkit for all our experiments.
CNN-LSTM [38]	ReLU	Cross Entropy	Adam	Accuracy: 92%	NVIDIA GeForce RTX 2080 Ti GPU (11GB) with the CUDA toolkit and cuDNN libraries installed on a Windows 10 Intel 64-bit OS for GPU-accelerated
DNN-RNN [3]	softmax, tanh	Cross Entropy	RMSprop	Accuracy: 50.2%	Python neural networks library, Keras, running on CUDA GPU.
BLSTM [68]	Tanh	Cross Entropy	Adam	Average: 68%	Experiments were performed on Google Colaboratory cloud environment.
Transformer [91]	Softmax	CTC Loss	Noam Optimizer	WER: 5.7%	Intel Xeon E5-2680 CPU, and 4 NVIDIA P100 GPUs.
TCAN [92]	Leaky ReLU	Cross Entropy	AdamW	Accuracy: 77.73%	Experiments were performed on Google Colaboratory cloud environment.
Word2Vec [96]	GeLU	Negative Likelihood Loss	Adam	Accuracy: 95%	Utilized the SpeechBrain toolkit for all our experiments.
BLSTM [9]	Softmax	CTC Loss	Adam	WER: 4% (improvement)	Multiple GPUs speed up training.
CNN-Transformer [114]	Sigmoid	Cross Entropy Loss + SDC Loss	Adam	WER: 6.5%	Model trained on 8 GPUs.
DBN [18]	Sigmoid	MSE	Dragonfly-Bird Swarm Optimization	Accuracy: 96.96%	MATLAB on a Windows 10 OS with an Intel i3 processor and 4GB RAM.

system. The EER is determined as follows:

$$EER = \frac{FAR + FRR}{2} \quad (5)$$

6) RECEIVER OPERATING CHARACTERISTIC

For varying values, the ROC curve displays the FAR as a function of the FRR. The ROC curve is a graphical representation of the performance of the classification

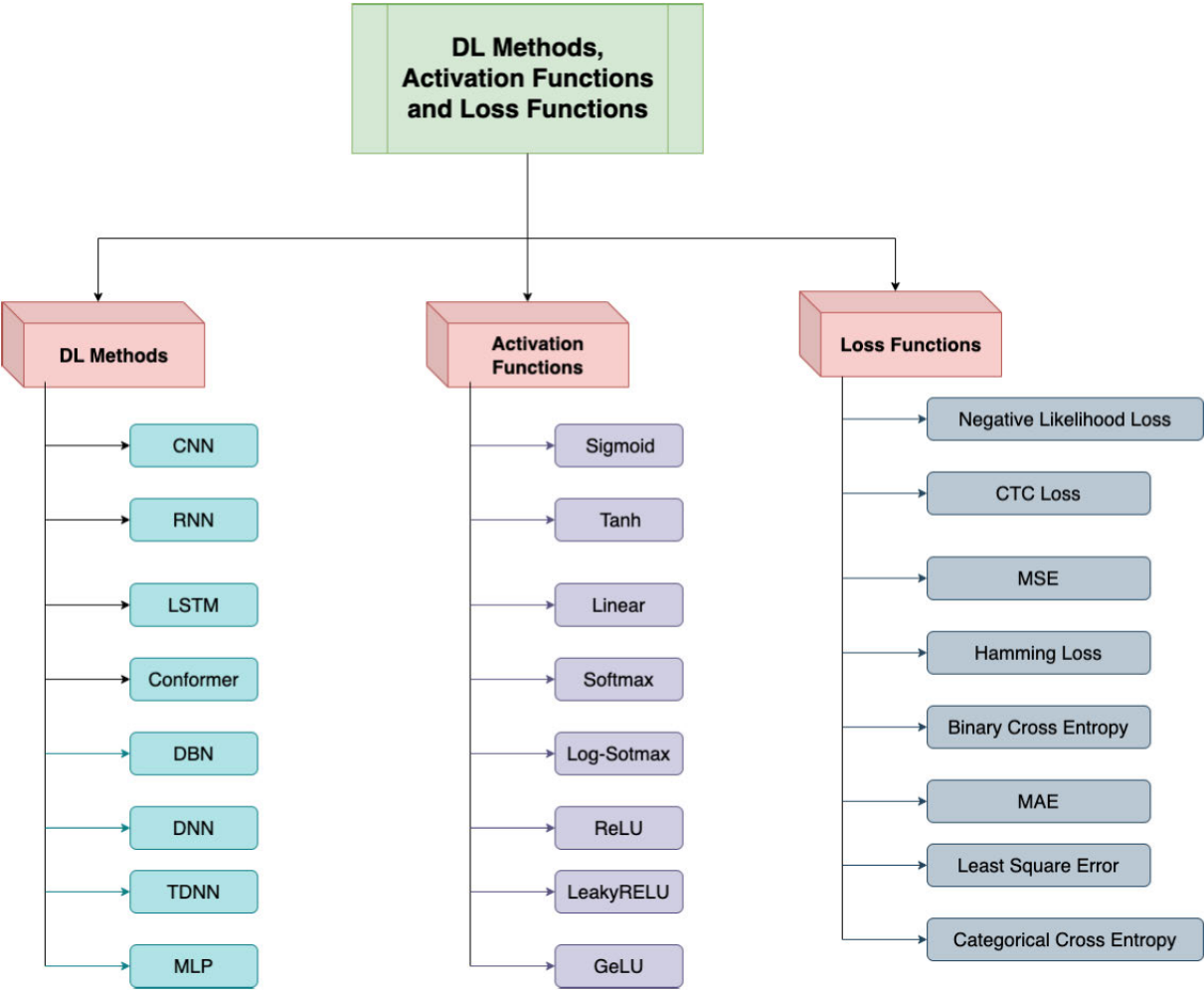


FIGURE 8. Deep learning methods, loss functions, and activation functions taxonomy for accent recognition.

algorithm to the discriminating threshold. The size of the training database and the decision threshold used to calculate the scores affect the FAR and FRR.

7) WORD ERROR RATE

For example, it is the ratio of transcription errors to the total number of words uttered. A lower WER in voice-to-text suggests higher speech recognition accuracy and a 20% WER indicates that the transcript is 80% correct.

D. DATASETS USED FOR THE ACCENT RECOGNITION

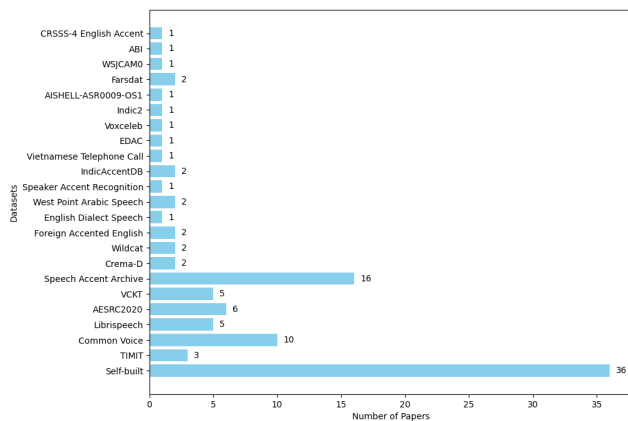
The selection of databases, as in many other technologies, is critical. A vast number of datasets have been utilized in the accent recognition field. Accent classification is a research domain within language and speech processing that centers on identifying and recognizing accents present in spoken language. Therefore, the dataset plays a vital role in enhancing the performance of accent detection systems. Several databases with various qualities that may be found either paid or free are utilized in this industry. Different

datasets can be used during the training process by utilizing them. Numerous types of datasets are accessible in this field, and Table 8 lists the databases that are employed in accent recognition. Figure 9 depicts the frequency of the datasets used for accent identification in the reviewed studies.

Accent recognition ability can be influenced by the language used, as different languages exhibit distinct auditory characteristics. While the majority of accent recognition datasets listed in Table 8 are designed for English, this paper also discusses datasets covering other languages. The frequency distribution of languages utilized for accent recognition in the reviewed studies is illustrated in Figure 10. The study predominantly focuses on accent recognition in widely spoken languages, particularly English, as accented speech recognition has been extensively studied for English due to several key factors: (1) its global prevalence, (2) the availability of large-scale datasets, (3) the diversity of accents influenced by non-native speakers, (4) its importance in international communication, and (5) its widespread application in speech-based technologies. However, our

TABLE 7. Frequency of evaluation metrics for accent recognition. ×: metric not used; ✓: metric used.

References	Accuracy	Precision	Recall	F1 Score	EER	ROC	WER
[6], [50], [24], [52], [26], [42], [17], [10], [72], [36], [37], [27], [5], [40], [22], [43], [77], [65], [70], [75], [76], [81], [92], [51], [119], [82], [115], [49], [116]	✓	×	×	×	×	×	×
[58]	×	✓	✓	×	✓	×	×
[72]	✓	✓	✓	×	×	×	×
[13], [21], [38], [45], [66], [113]	✓	✓	✓	✓	×	×	×
[7], [20], [60]	✓	✓	✓	✓	×	✓	×
[61]	✓	✓	✓	✓	×	×	×
[18], [62]	✓	×	×	✓	×	×	×
[89], [114], [118], [12], [9], [94], [97]	×	×	×	×	×	×	×
[63]	×	✓	✓	✓	×	×	×
[28], [14], [90], [80], [93]	✓	×	×	×	✓	×	×
[15], [79]	✓	×	×	×	×	✓	×
[90], [87]	✓	×	×	×	×	✓	×
[29]	×	×	×	×	✓	×	×
[8], [39], [33], [73]	✓	×	×	×	✓	×	×
[118], [12], [91], [114], [9]	×	×	×	×	×	×	✓

**FIGURE 9.** The frequency of datasets in the research reviewed.

primary focus is on different English accents, including Indian, German, Italian, and Chinese, among others. This explains why Figure 10 is particularly relevant for English.

VI. ACCENT RECOGNITION FOR INDIAN LANGUAGES

India has 122 primary languages and 1599 other languages, based on the 2001 census of India. According to the People's Linguistic Survey of India, India has the second most languages (780). The performance of European speech recognition engines has improved significantly [120]. However, most Indian languages are still in the early stages of the ASR system development. Languages such as Punjabi, Assamese,

Kashmiri, Dogri, and others have not been researched much because no standard speech dataset of Indian languages and dialect variances exists. Some languages still lack the minimum speech dataset necessary for system development.

Reference [121] Proposes CNN, a DL-based approach for accent classification, has a 98.66% accuracy utilizing the MFCC and Mel spectrograms features. The paper [101] proposed a technique for detecting accents in Telugu speech with an accuracy of 72%, where the authors collected speeches from three different Telugu accents and extracted data such as syllable lengths, pitch, and energy contours. Table 9 shows the accented speech recognition for some Indian languages.

VII. FUTURE TRENDS AND RESEARCH GAPS

RQ5: What are the future trends and research gaps in accent Recognition?

Accent identification is an area in which researchers have shown extensive interest and focus. This is regarded as an issue in the field of AI. Accent recognition can be used in mobile devices, desktop computers, and call centers as virtual assistants. However, there are other challenges and concerns that researchers have attempted to investigate.

Recent studies in 2022 and 2023 suggest potential future study topics for accent identification. Most of these studies have concentrated on improving the offered methodologies and boosting the accuracy of accent identification in various

TABLE 8. Frequently used datasets for accent recognition.

Dataset	Description	References
Speech Accent Archive	The Speech Accent Archive offers an extensive collection of spoken recordings from individuals across the globe, each reciting an identical passage in English. It presents a diverse range of linguistic accents and serves as a valuable resource for scholars investigating variations in pronunciation and speech patterns.	Speech Accent Archive
TIMIT	The TIMIT dataset comprises speech recordings from a total of 630 individuals, representing eight distinct American English dialects. Every participant recites ten sentences designed to be phonetically diverse, offering essential resources for research in both phonetics and ASR.	TIMIT Corpus
LibriSpeech	The LibriSpeech dataset comprises nearly 1,000 hours of 16kHz English read speech, sourced from LibriVox audiobooks. Its extensive size and high quality make it a standard choice for ASR system training and evaluation.	LibriSpeech Corpus
Common Voice	Mozilla's Common Voice is a publicly available dataset with multiple languages, built through global volunteer contributions. It includes diverse speech samples to enhance accessibility and representation in accented speech recognition technology.	Common Voice Dataset
IndicAccentDB	IndicAccentDB is a dataset centered on English speech from individuals with diverse Indic language backgrounds. It supports accent identification and enhances speech technology adaptation to accommodate accent diversity across India's linguistic spectrum.	Indic TTS and ASR Corpora
VoxCeleb	VoxCeleb is an extensive dataset featuring celebrity voices, derived from YouTube clips. It comprises more than a million speech samples across numerous speakers, serving as a valuable asset for research in speaker identification, verification, and accent evaluation.	VoxCeleb Dataset
EDAC	The Emotional Diverse Actors Database (EDAC) comprises a collection of speech samples performed by actors, encompassing various emotional expressions. This dataset facilitates studies in affective computing, emotion classification, and the synthesis of expressive speech, capturing an extensive spectrum of emotional states.	Not available
West Point Arabic Speech	Created by the United States Military Academy, the Arabic Speech Corpus from West Point comprises both Modern Standard Arabic and multiple dialects, serving as a crucial asset for research in accented speech recognition and language processing.	West Point Arabic Speech
CREMA-D	The Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D) comprises video and audio recordings of performers expressing various emotions. This dataset serves as a valuable asset for research in affective computing, multi-modal analysis, and emotion recognition.	CREMA-D Dataset
VCTK	The VCTK Corpus comprises recordings of speech from 110 individuals who speak English with diverse accents. Collected at the University of Edinburgh, this dataset is widely utilized in studies related to voice transformation, speech generation, and linguistic accent analysis.	VCTK Corpus
Farsdat	Farsdat serves as an extensive speech dataset for Persian (Farsi), containing recordings of native Iranian speakers. Covering multiple dialects, it plays a crucial role in advancing research on speech recognition, synthesis, and linguistic studies in Persian.	Farsdat Information

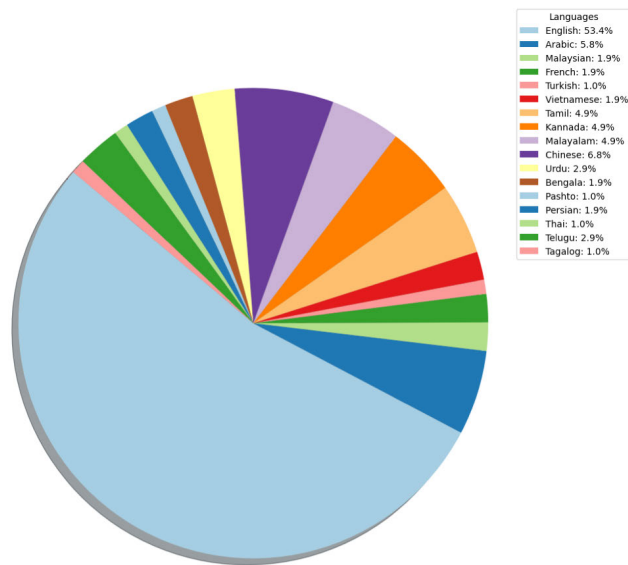


FIGURE 10. Frequency distribution of languages used in the research reviewed.

situations. Table 10 presents the future direction of some of the publications on accent recognition, which illustrates the goals and directions that researchers intend to examine in the field of accent identification.

This review explored related works in the area of accent recognition. The major research gaps and the potential research directions are listed below.

A. DATASET

Speech database design is a key step in building an accent recognition system. A rich speech database is essential for creating and developing accent recognition systems suited to specific applications. Most of the available datasets were collected through the direct interaction of speakers without considering external noise. There has been significant progress in the accent identification of the English language. In most Indian languages, research on the development of automatic accent identification systems is still in its early stages. This is because Indian languages and their dialectal variations lack a standard speech dataset. Only a few studies have considered automatic accent recognition of many available Indian languages. Since a greater proportion of the Indian population needs to use their respective native languages for communication, there is a need for better ASR systems. If an appropriate recognition approach is not supplied to the automatic accent recognition system, a large-scale voice corpus might increase the computing complexity and significantly reduce the performance during testing.

Despite the advancements in DL-based speech recognition, the development of accented speech recognition systems for low-resource languages remains a significant challenge due to the lack of extensive and well-structured speech datasets. Most existing studies have focused on high-resource languages, where large-scale corpora are readily available, while low-resource languages, still suffer from the absence of

standardized speech databases. This gap in data availability limits the ability of DL models to effectively learn the intricate variations in accents, dialects, and environmental conditions. Without sufficient and diverse datasets, even state-of-the-art DL architectures struggle with generalization, leading to poor recognition accuracy. Therefore, a critical research gap exists in the creation of large-scale, high-quality speech datasets that encompass diverse speakers, speaking styles, and real-world noise conditions. Addressing this challenge requires comprehensive data collection strategies, including crowd-sourcing, field recordings, and collaboration with native speakers, to ensure the development of robust and efficient accent speech recognition systems for low-resource languages.

B. IMPROVING ACCENT RECOGNITION MODELS

While DL approaches have significantly advanced accent detection, challenges remain in developing models that can reliably recognize accents across speakers from diverse linguistic backgrounds. Current models often struggle with variations in pronunciation, intonation, and articulation influenced by native language interference. Several techniques have been explored to enhance model performance, including multi-accent training for low-resource languages, which helps improve recognition in underrepresented accents. Additionally, audio data augmentation can introduce variations that improve model generalization, while incorporating accent diversity in training datasets ensures broader adaptability. Fine-tuning pre-trained models has shown promise in leveraging existing knowledge to enhance accent classification. However, factors such as background noise, inconsistent speech patterns, and variations in speaker demographics can negatively impact recognition accuracy. Future research should focus on refining these techniques, optimizing model robustness, and addressing the limitations that hinder reliable accent detection across diverse linguistic groups.

C. DEPLOYING ACCENT RECOGNITION SYSTEMS IN REAL-WORLD ENVIRONMENTS

While the literature extensively evaluates the performance of accent recognition models under controlled laboratory conditions, there remains a significant gap in understanding how these models perform in real-world scenarios. Most studies focus on benchmark datasets with clean and well-annotated speech samples, often neglecting challenges such as background noise, speaker variability, microphone differences, and temporal fluctuations in speech patterns. These factors can substantially impact model robustness and generalization, limiting their practical applicability. Future research should explore techniques like domain adaptation, real-world data augmentation, and transfer learning to bridge this gap. Additionally, deploying models in real-world environments and evaluating their performance across diverse acoustic conditions will provide deeper insights into their effectiveness and reliability for real-world applications.

TABLE 9. Accented speech recognition for different indian languages.

Reference	Language	Feature Used	Accent Technique	Recognition	Performance
[122]	Bangla	MFCC	GMM		Accuracy: 84%
[105]	Marathi	DWT	ANN		Accuracy: 60%
[37]	Telugu	MFCC	GMM		Accuracy: 91%
[123]	Hindi	MFCC	GMM		Accuracy: 95%
[124]	Gujarati	MFCC, Real Cepstral Co-efficient	ANN		Accuracy: 94%
[125]	Assamese	MFCC	HMM, VQ, I-vector		Accuracy: 99.99%
[126]	Tamil	MFCC	HMM		WER:
[127]	Hindi	MFCC	HMM		Accuracy: 87%
[128]	Malayalam	MFCC	HMM		Accuracy: 94.8%
[121]	Kashmiri	Mel Spectrograms, MFCC	CNN		Accuracy: 99%
[101]	Telugu	MFCC, Energy, Formant	NNC		Accuracy: 72%
[116]	Telugu	MFCC, Pitch Chroma, Tonal Power Ratio, Spectral Flux	DBN		Accuracy: 93%
[56]	Tamil	Acoustic Mel Cepstral and Visual Features	HMM		Accuracy: 96%

D. GENERATIVE AI FOR ACCENT RECOGNITION

GANs can be used to train a network on a large collection of voice samples from various accents. The generator network learns to produce accent-invariant speech samples, whereas the discriminator network learns to differentiate between original and generated speech samples. Accent-invariant representations can be extracted from voice samples by using GANs. These representations may be fed into a voice recognition system, allowing them to be effectively generalized across accents. Therefore, GANs for accent-invariant representation learning are of great research interest. This helps create a unified end-to-end speech recognition system that generalizes well across accents.

E. EXPLAINABLE AI (XAI) FOR ACCENT RECOGNITION

DNNs are a potential approach to recognizing accents. In contrast to other types of data, such as natural images, audio data have various sensible and non-obvious representations that can be used as inputs to these models. Because of their black-box character, the influence of diverse input representations has been studied primarily through classification performance. XAI was developed to understand the underlying classification techniques of models trained on various input representations. Therefore, there is a need to develop new XAI algorithms and techniques to provide relatable explanations for audio prediction. The insights gained from the XAI-based model not only assist us in understanding the model's basic reasoning processes, but they may also influence the construction of future models specialized in audio applications and aligned with human expectations. Therefore, an accent recognition model with

explainable tools can provide a better insight into the development of more accurate ASR systems.

F. CROSS-LINGUAL ACCENT CLASSIFICATION

Most current accent recognition techniques are designed using models trained with speech data from speakers who all use the same language. This approach works well in controlled environments but falls short in real-world settings where accents may vary greatly. The challenge arises because these models struggle to accurately identify accents when confronted with speakers of different languages than those present in their training data. This limitation can hinder effective communication and pose obstacles in multilingual environments. Therefore, there is a significant need to advance ML models that can reliably recognize accents across various languages. Such advancements would greatly enhance communication and support language learning by providing more accurate accent detection in diverse linguistic landscapes.

G. MULTI-MODAL LEARNING FOR ACCENT RECOGNITION

Traditional accent recognition models primarily rely on acoustic features extracted from speech signals. However, integrating multi-modal information, such as visual cues, linguistic context, and articulatory features, can significantly enhance performance. For instance, lip movement analysis aids in distinguishing accents, as phoneme articulation varies across languages. Future research can focus on DL architectures that fuse speech and lip-reading features, transformer-based multi-modal models like Audio-Visual BERT for improved generalization, and the integration of

TABLE 10. Future directions for accent recognition in reviewed studies.

References	Year	Proposal for Future Directions
[52]	2023	In the future, we will expand this research with a larger and more diversified database and investigate additional feature optimization approaches such as PCA and PSO.
[14]	2022	A variety of cues based on known facts regarding how the accent is recognized by humans, including the variables that characterize speakers, can be investigated. In the future, greater knowledge of the acoustic-phonetic characteristics of accents may lead to the synthesis of regionally accented speech.
[23]	2022	In the present study, inputs were provided in the form of numbers, that is music features. In the future, we may use spectrogram amplitude graphs of certain audio to input these graphs/images into the model and examine other aspects using the STFT.
[59]	2022	Gender-accent hybrid techniques for the selection of mini-batches have been developed to improve the performance in voice recognition tasks. In the future, additional features should be investigated for the selection of mini-batch samples for accent voice recognition.
[62]	2022	Future studies will investigate the bias of different classifier designs towards specific social backgrounds as well as the influence of LPF cutoff frequencies for augmentation.
[45]	2022	Because speech data contain a time-dependent recurrence connection, sophisticated recurrence-based models for accent classification, such as GNN and Transformer-based models, can be employed in the future.
[40]	2019	In the future, we can delve deeper into ELM by including other layers, such as the "ML-ELM" and "H-ELM" approaches. Furthermore, integrating the RNN with PCA can improve the overall accuracy of the model.
[50]	2022	Further research should employ an intermediate classifier to predict a speaker's L1 language group before categorizing a specific accent and utilizing a dataset with a range of spoken materials.
[119]	2022	In the future, we will undertake experiments on more Vietnamese speech data and lower the model size to accommodate low-resource devices. In addition, we will investigate the efficacy of pre-trained unsupervised models for Vietnamese accent identification.
[12]	2022	Because this study concentrated on South Asian English dialects, there was a small increase in CER and WER for English speakers. The method will be further scaled to target other languages of South Asia such as Hindi, Urdu, Bengali, and others, with more robust data corpora and improved accuracy, as well as concurrent training of both pipelines.
[63]	2022	Future research should examine the use of RNNs and CNNs as classifiers, in conjunction with improved data augmentation approaches. Furthermore, a larger dataset with geographically closer accents would undoubtedly aid in training NNs in fine-tuning pre-trained speech models.
[82]	2022	Future work on the ASR will be based on a two-stage method. In the first stage, the accent is recognized and utilized as an input to the later stage, with the MFDWC or SSC properties being relevant for the first stage.
[17]	2021	New approaches, such as direct waveform processing with LEAF [129] to lower package loss from the Fourier transform, can also be used to improve accuracy.
[64]	2021	In future investigations, a hybrid sample selection strategy for mini-batch samples based on successful mini-batch sample selection methods should be provided, and the effect of sample selection strategies for mini-batches with a language model should be investigated.
[9]	2022	In future studies, we will apply a temporal convolution network to enhance accent Mandarin ASR systems with minimal data. In addition, we examined the performance of the accent-dependent bottleneck features.
[10]	2020	Such accent categorization algorithms may also be utilized as preprocessing add-ons to ASR systems before transcription. Therefore, we recommend more studies on accent classification in many languages
[42]	2019	Feature reduction methods such as PCA will be used in future work to minimize feature dimension. This will be refined to enhance the outcomes. Experimenting with other numbers of coefficients and other wavelet families to see the recognition outcome would be interesting.
[68]	2019	In the future, we want to enhance accent categorization using tone information from ASR. In addition, we will look at using the accent recognizer to detect accented training data for offline AM adaption.
[22]	2018	In the future, we will examine other terms with noticeable differences between diverse accents and dialect groups.
[30]	2016	Because the data for each accent are limited, a universal recognizer based on the RBM can be explored instead of typical GMMs for each accent.

textual information, such as phoneme transcripts, alongside audio features to enhance recognition accuracy.

H. FEW-SHOT AND ZERO-SHOT LEARNING FOR LOW-RESOURCE ACCENTS

Accent recognition faces challenges due to data scarcity, particularly for underrepresented languages and dialects. Few-shot learning and zero-shot learning offer solutions by enabling models to generalize to new accents with minimal or no labeled data. Meta-learning approaches, such as model-agnostic meta-learning, further enhance adaptability to novel accents. Future research could focus on leveraging contrastive learning to improve accent generalization, utilizing pre-trained embeddings like Wav2Vec 2.0 for few-shot learning, and developing zero-shot models capable of classifying accents without explicit training data.

I. SELF-SUPERVISED LEARNING FOR ACCENT REPRESENTATION

Self-supervised learning methods, including Wav2vec 2.0, and Whisper, have achieved state-of-the-art performance in various speech tasks by learning meaningful representations from unlabeled speech data. These models are particularly beneficial for accent recognition, where labeled data is often limited. Future research directions include training self-supervised learning models on accent-rich speech corpora, fine-tuning large self-supervised learning models for accent recognition, and exploring contrastive learning techniques to enhance accent feature extraction.

VIII. CONCLUSION

This study conducted a comprehensive review of 103 conference papers and journal articles on accent recognition, published between 2015 and 2023 across four major databases: ACM Digital Library, Scopus, IEEE Xplore, and Science Direct, exploring the application of ML and DL techniques. It systematically detailed the research methodology, including research questions and search strategies for paper selection. Also, this work analyzed various preprocessing techniques, feature extraction methods, and accent recognition models encompassing traditional and DL-based frameworks. Examining the characteristics of audio signals provided insights into state-of-the-art DL approaches, including activation and loss functions. Furthermore, commonly used datasets for accent recognition were reviewed, and the challenges posed by auditory variations were discussed. The study concluded by identifying key research gaps and proposing future directions to advance accent recognition systems, offering valuable guidance for researchers to address existing challenges and explore innovative methodologies in the field.

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